# Sebastián David Pinzón Zambrano

Junior Data Engineer · Reproducible data pipelines and applied ML

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## PROFILE

I build data pipelines and machine learning models in Python with one simple rule: anyone should be able to clone the repository, run the tests and get the same result. My public work uses financial market data: API ingestion, validation, Parquet and Delta Lake storage, PySpark medallion pipelines, leak-free feature engineering and models served with FastAPI. Every project has tests, most have CI and Docker, and each README states what works and what doesn't. Self-taught, with management experience and proven discipline in demanding environments. I am looking for my first role as a junior Data Engineer or on a data/ML team.

## TECHNICAL SKILLS

**Languages and data:** Python, SQL, PostgreSQL, pandas, NumPy, Parquet
**Data engineering:** PySpark, Delta Lake, medallion architecture, ETL, data validation
**ML:** scikit-learn, XGBoost, SHAP, Optuna, Hugging Face, walk-forward validation
**Engineering and delivery:** Git, GitHub Actions (CI), Docker, pytest, FastAPI, Pydantic, Streamlit

## SELECTED PROJECTS (GITHUB.COM/ARES-INFENUS)

### SparkFlow Delta Pipeline

PySpark · Delta Lake · Docker · GitHub Actions

Batch medallion pipeline (Bronze, Silver, Gold) for rule-based fraud detection on synthetic banking transactions. Generator configurable up to 100M rows (run at 100k), partitioning, ZSTD compression, Z-ORDER and time travel; 24 tests and CI.

github.com/Ares-Infenus/SparkFlow_Delta_Pipeline

### Alpha Feature Engineering

pandas · scikit-learn · statsmodels · pytest

Turns daily OHLCV data into 77 stationary features per asset, designed to avoid look-ahead bias. Validation layer, scikit-learn-compatible transformers and TimeSeriesSplit diagnostics; 42 tests, including anti-leakage checks.

github.com/Ares-Infenus/alpha-feature-engineering-timeseries

### BTC/USDT Direction Predictor

XGBoost · Optuna · SHAP

Experiment on 28,000+ hourly Binance candles with 15-fold walk-forward validation. ROC-AUC 0.543 ± 0.028: reported as a weak signal, not a trading strategy; 98 tests.

github.com/Ares-Infenus/xgboost-financial-predictor

### InsightFlow NLP

FastAPI · Hugging Face · spaCy · Docker

REST API with 8 endpoints (sentiment, entities and topics; batches of up to 500 texts) and a Streamlit dashboard; ~170 ms per sentiment request; 44 tests and CI.

github.com/Ares-Infenus/InsightFlow_NLP

### Penguin Species API

FastAPI · Docker · Render

Deployed ML service with a live demo, strict input validation and golden-dataset regression tests; 27 tests and CI.

github.com/Ares-Infenus/penguin-ml-api · Live demo: penguin-ml-api-bmrr.onrender.com/docs

## EXPERIENCE

**Junior Machine Learning Contributor (volunteer) — Omdena** Oct 2026 – present

- “GeoData Scout” project with Humanitarian OpenStreetMap Team (HOT fAIr): researching and preparing geospatial datasets for AI base models.

**Administrative and Operations Assistant — Colombian National Army (military service)** Aug 2025 – present

- Official documentation and supply for platoons; drafting and records control of reports and incident logs.
- Administrative support for a platoon in a high-discipline, high-pressure environment.

**Self-directed training in data science and ML — Independent study** Aug 2024 – Jul 2025

- Programs from Stanford, DeepLearning.AI, Duke, IBM, Google, Johns Hopkins and the University of Michigan in Python, SQL, MLOps and deep learning.

**General Manager — Imperio del Plástico, Valledupar** Jun 2023 – Jun 2024

- Ran the business day to day: sales, margins and monthly results in Excel, with dashboards and charts.
- Sales forecasting with trend lines and moving averages over historical data; Python and SQL to clean and transform accounting and sales data.
- Launched a grill-manufacturing line that was profitable in its first quarter.

## EDUCATION AND CERTIFICATIONS

- **Universidad Nacional de Colombia** — Systems Administration (Administración de Sistemas Informáticos), 2019–2023. Seven semesters completed; programme not finished.
- **14 Coursera certifications**, including Machine Learning (Stanford), TensorFlow Developer and Deep Learning (DeepLearning.AI), MLOps (Duke), Google Data Analytics, IBM Data Science, Applied Data Science with Python and PostgreSQL for Everybody (University of Michigan).
- **Top score in the national Saber 11 exam** — Municipal Education Secretariat, 2018.

## LANGUAGES

Spanish: native. English: basic (A2), actively studying with a B2 target.